THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · HARD LEVEL UNIT TEST SET 2

Chapters 8–11: **Introduction to Trigonometry, Some Applications of Trigonometry, Circles, Areas Related to Circles** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A sector of a circle of radius 42 cm has central angle 45°. The perimeter of the sector is: 1

(a) 159 cm **(b)** 33 cm **(c)** 117 cm **(d)** 75 cm

2. $\dfrac{\sin^2 25^\circ + \sin^2 65^\circ}{\cos^2 35^\circ + \cos^2 55^\circ} =$ 1

(a) 1 **(b)** $\frac{1}{2}$ **(c)** 2 **(d)** 0

3. From the top of a building 3 m high, the angle of elevation of the top of a tower is 30° and the angle of depression of its foot is 60°. The height of the tower is: 1

(a) $\sqrt{3}$ m **(b)** 1 m **(c)** $4\sqrt{3}$ m **(d)** 4 m

4. In $\triangle ABC$, right-angled at B, $AB = 45$ cm and $BC = 60$ cm. A circle is inscribed in the triangle. The radius of the inscribed circle is: 1

(a) 37 cm **(b)** 15 cm **(c)** 30 cm **(d)** 16 cm

5. An arc of length 22 cm subtends an angle at the centre of a circle of radius 42 cm. The angle subtended by the arc at the centre is: 1

(a) 150° **(b)** 30° **(c)** 15° **(d)** 60°

6. The value of $\tan 10^\circ \tan 20^\circ \tan 70^\circ \tan 80^\circ$ is: 1

(a) 2 **(b)** $\tan 45^\circ + 1$ **(c)** 0 **(d)** 1

7. Two points on level ground lie on opposite sides of a tower of height 9 m, in line with its foot. The angles of elevation of the top of the tower from the two points are 45° and 60°. The distance between the two points is: 1

(a) $9 + 9\sqrt{3}$ m **(b)** 12 m **(c)** $9 + 3\sqrt{3}$ m **(d)** $9 - 3\sqrt{3}$ m

8. A point P is 85 cm from the centre O of a circle of radius 77 cm. The length of the tangent drawn from P to the circle is: 1

(a) 1296 cm **(b)** 8 cm **(c)** 162 cm **(d)** 36 cm

9. A square is inscribed in a circle of radius 35 cm. The area of the circle outside the square is: 1

(a) 3850 cm² **(b)** 1400 cm² **(c)** 2625 cm² **(d)** 2450 cm²

10. If $\sin\theta = \frac{12}{13}$ then $\sin^2\theta + \cos^2\theta + \tan^2\theta =$ 1

(a) $\frac{25}{169}$ **(b)** $\frac{144}{169}$ **(c)** $\frac{144}{25}$ **(d)** $\frac{169}{25}$

11. A person looks at the top of a vertical pole from a point on the ground. The top of the pole is $4\sqrt{3}$ m above the level of the person's eyes and the eyes are 12 m horizontally from the pole. The angle of elevation of the top of the pole is: 1

(a) 60° **(b)** $30\sqrt{3}^\circ$ **(c)** 45° **(d)** 30°

12. AB is a chord of a circle with centre O and PA, PB are tangents. Then the line OP is: 1

(a) parallel to AB **(b)** equal to AB

(c) the perpendicular bisector of AB **(d)** a tangent to the circle

13. **Assertion (A):** The area of a sector of a circle of radius 7 cm and angle 90° is $38.5\ \text{cm}^2$ (use $\pi = \frac{22}{7}$). 1

Reason (R): The area of a sector of angle θ is $\dfrac{\theta}{360^\circ} \times \pi r^2$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

14. **Assertion (A):** $\tan 45^\circ = 1$. 1

Reason (R): $\tan\theta = \dfrac{\cos\theta}{\sin\theta}$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

15. **Assertion (A):** A tower $10\sqrt{3}$ m high casts a shadow 10 m long when the angle of elevation of the Sun is 60°. 1

Reason (R): $\tan 60^\circ = \dfrac{1}{\sqrt{3}}$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. A flagstaff stands on the top of a building 9 m high. From a point on level ground the angles of elevation of the bottom and the top of the flagstaff are 45° and 30° respectively. Find the length of the flagstaff. 2

17. PA and PB are tangents from P to a circle, each of length 9 cm. A third tangent touches the circle at E on the minor arc AB and meets PA at C and PB at D. Find the perimeter of $\triangle PCD$. 2

18. A wheel has radius 35 cm. Find the number of complete revolutions it makes to cover 110 m. 2

19. If $x = a\sec\theta$ and $y = b\tan\theta$ with $a = 2$, $b = 3$, express $\dfrac{x^2}{4} - \dfrac{y^2}{9}$. 2

20. A car moves at a uniform speed towards the foot of a vertical tower. The angle of elevation of the top of the tower changes from 30° to 60° in 6 minutes. Find the further time the car takes to reach the foot of the tower. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. A quadrant of a circle of radius 63 cm is cut from a square of side 63 cm, with the centre of the quadrant at one corner of the square. Find the area of the remaining part of the square. 3

22. If $\tan\theta + \cot\theta = 3$, find $\tan^2\theta + \cot^2\theta$. 3

23. Two points on level ground lie on opposite sides of a tower of height 6 m, in line with its foot. The angles of elevation of the top of the tower from the two points are 45° and 60°. Find the distance between the two points. 3

24. Tangents QA and QB are drawn from an external point Q to a circle, and $QA = (21x + 5)$ cm and $QB = (13x + 29)$ cm. Find the length of QA. 3

OR

The minute hand of a clock is 42 cm long. Find the area swept by the minute hand in 5 minutes. 3

25. Prove that: $\sec^2\theta + \csc^2\theta = \sec^2\theta\,\csc^2\theta$. 3

OR

A building is $20\sqrt{3}$ m high. From the top of the building the angle of elevation of the top of a tower is 30° and from the foot of the building it is 60°. Find the height of the tower and its distance from the building (in this order). 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. **Case study.** A vertical tower stands on level ground. From a point P on the ground, 20 m from the foot of the tower, the angle of elevation of its top is 60°. (Use $\sqrt{3} = 1.73$.) 5

(i) Name the trigonometric ratio that relates the height of the tower to the distance of P from the foot of the tower.

(ii) Write the value of $\tan 60^\circ$.

(iii) Find the height of the tower.

(iv) The observer moves back until the angle of elevation becomes 30°. Find the new distance from the foot of the tower and how far the observer moved.

27. Tangents PA and PB are drawn from a point P, 75 cm from the centre O of a circle of radius 72 cm. The length of each tangent is 21 cm. Find the length of the chord AB. 5

OR

Two concentric circles have radii 21 cm and 14 cm. A sector of angle 36° is marked in both. Find the area of the region between the two arcs within the sector. 5

END OF QUESTION PAPER — **Total: 50 Marks**